

Software Requirements Specification (SRS)

E-Commerce Management System (Console-Based C++ Application)

1. Introduction

1.1 Purpose

The purpose of this document is to describe the **functional** and **non-functional requirements** of a **Console-Based E-Commerce Management System** implemented using **C++**.

The system allows management of **products**, **customers**, and **orders**.

1.2 Scope

The system enables:

- Adding products
- Adding customers
- Creating orders
- Adding products to orders
- Calculating order totals
- Displaying all stored data

This system is intended for **educational and learning purposes** to demonstrate **Object-Oriented Programming (OOP) principles** in C++.

1.3 Definitions

- **Product:** An item available for purchase.
- **Customer:** A user who places orders.
- **Order:** A purchase request made by a customer.
- **OrderItem:** A specific product and quantity within an order.

2. Overall Description

2.1 Product Perspective

This is a **standalone console application**.

No database or external API integration is used.

All data is stored **in memory** using **STL vectors**.

2.2 Product Features

The system includes:

- **Product Management**
- **Customer Management**
- **Order Management**
- **Order Total Calculation**
- **Displaying System Data**

2.3 User Classes

- **System Admin (Console User)**
 - Can add products
 - Can add customers
 - Can create and manage orders

2.4 Operating Environment

- **Language:** C++
- **Compiler:** Any standard C++ compiler (GCC, Clang, MSVC)
- **Platform:** Windows / Linux / macOS
- **Interface:** Console (CLI)

2.5 Constraints

- **No persistent storage** (data lost after program termination)
- No authentication system
- No graphical interface
- **Single-user system**

3. Functional Requirements

3.1 Product Management

- **FR-1: Add Product**
 - The system shall allow adding a new product with:

- ID (**auto-generated**)
- Name
- Price
- Stock quantity

- **FR-2: Display Product**

- The system shall display:
 - Product ID
 - Name
 - Price
 - **Available stock**

3.2 Customer Management

- **FR-3: Add Customer**

- The system shall allow adding a new customer with:
 - ID (**auto-generated**)
 - Name
 - Email

- **FR-4: Display Customer**

- The system shall display:
 - Customer ID
 - Name
 - Email
 - List of **order IDs**

3.3 Order Management

- **FR-5: Create Order**

- The system shall:
 - Create a new order for a **given customer ID**

- Assign a **unique order ID**
 - Link order to the customer
- **FR-6: Add Product to Order**
 - The system shall:
 - Add a product to an **existing order**
 - Store product ID and quantity
 - **Validate stock availability**
- **FR-7: Calculate Order Total**
 - The system shall:
 - Multiply **product price × quantity**
 - Sum all items
 - **Store total** in the order
- **FR-8: Display Order**
 - The system shall display:
 - Order ID
 - Customer ID
 - List of products
 - Quantities
 - **Total price**

3.4 System Display

- **FR-9: Display All Data**
 - The system shall:
 - Display **all products**
 - Display **all customers**
 - Display **all orders**

4. Non-Functional Requirements

4.1 Performance

- The system should handle at least **100 products** and **100 orders** efficiently.

4.2 Usability

- Console output should be **clear and readable**.
- Data should be **properly formatted**.

4.3 Reliability

- The system must validate:
 - **Product existence**
 - **Customer existence**
 - **Order existence**
 - **Stock availability**

4.4 Maintainability

- Code must follow **OOP principles**:
 - **Encapsulation**
 - **Modularity**
 - **Separation of responsibilities**